



NORTH SYDNEY BOYS' HIGH SCHOOL
2009 Year 11 Assessment Task 3

MATHEMATICS

(Extension 1)

General instructions

- Working time – 60 minutes.
- Write on your own A4 paper.
Each question is to commence on a new page.
- Write using blue or black pen.
- Board approved calculators may be used.
- All necessary working should be shown in every question (Insufficient/illegible working may cause a deduction of marks).
- Attempt **all** questions.

Class teacher (please ✓)

- ☐ Mr Barrett
- ☐ Mr Lowe
- ☐ Mr Rezcallah
- ☐ Mr Trenwith
- ☐ Mr Ireland
- ☐ Mr Weiss
- ☐ Mr Lam

NAME: PAGES USED:

Marker's use only.

QUESTION	1	2	3	4	5	Total	%
MARKS	$\overline{12}$	$\overline{9}$	$\overline{13}$	$\overline{9}$	$\overline{11}$	$\overline{54}$	

Question 1 (12 Marks)	Commence a NEW page.	Marks
(a) Differentiate with respect to x , simplifying the answer fully.		
i. $f(x) = 3x^3 + x^2 - 12x + 4$.		2
ii. $y = (5x^2 - 4)^2$.		2
iii. $f(x) = \frac{3x}{x^2 - 4}$.		2
iv. $y = 5x^3\sqrt{x+2}$.		3
(b) Find the value(s) of x for which the tangent to the curve $y = \frac{5}{3}x^3 - 10x^2 + 6$ is parallel to the x -axis.		3

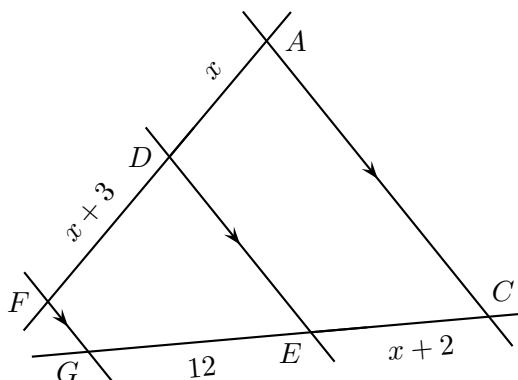
Question 2 (9 Marks)	Commence a NEW page.	Marks
(a) Shade in the region defined by		3
$y \leq \sqrt{4 - x^2} \quad \text{and} \quad y > x - 2$		
(b) AB is divided by Q externally in the ratio $2 : 3$. Find the co-ordinates of B if A is $(3, -5)$ and Q is $(-4, 7)$.		3
(c) Find the equation of the line through the point of intersection of the two lines given by the equations $x + 2y + 1 = 0$ and $2x - 3y + 5 = 0$ and through the point $(1, -3)$		3

Question 3 (13 Marks)

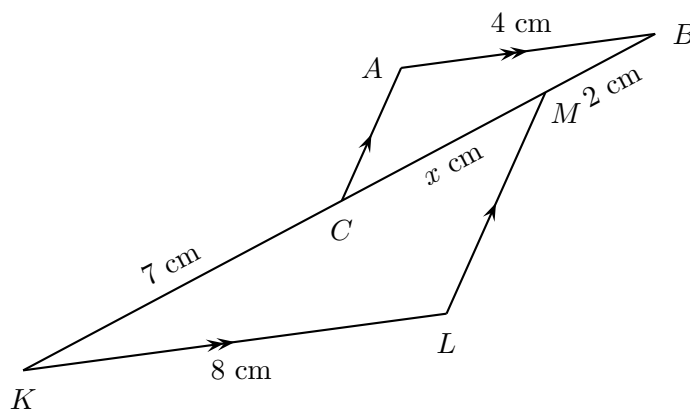
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Marks

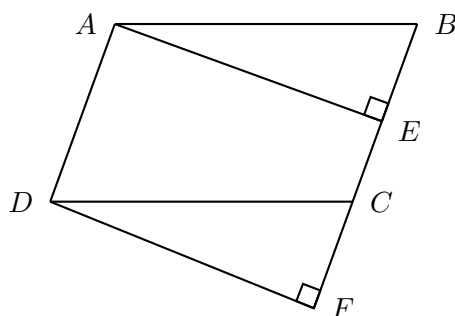
- (a) In the diagram, $DE \parallel AC$. Find the value(s) of x , showing full working.

3

- (b) In the diagram AB is parallel to KL and AC is parallel to ML . $AB = 4$ cm, $MB = 2$ cm and $KC = 7$ cm.



- Prove $\triangle ABC$ is similar to $\triangle KLM$. **3**
 - Find the value of x . **2**
- (c) $ABCD$ is a parallelogram. $\angle AEB = \angle DFC = 90^\circ$.



- Prove $\triangle ABE \equiv \triangle DCF$. **3**
- Show that $AEFD$ is a rectangle. **2**

Question 4 (9 Marks)

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Marks

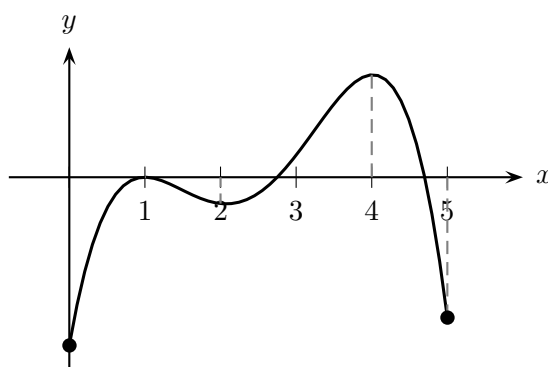
- (a) Given $\log_a 9 = 1.585$ and $\log_a 10 = 1.661$, find the following values correct to 3 decimal places.
- $\log_a 90$. **1**
 - $\log_a 0.9$. **1**
- (b) Find the value of $\log_9 10$, correct to 2 decimal places. **2**
- (c) Solve for x . **2**
- $$0.4^x \leq 5$$
- (d) Solve the equation $2\log_2(x-1) = \log_2(3x-3)$. **3**

Question 5 (11 Marks)

Commence a NEW page.

Marks

- (a) Evaluate the following limits, expressing your answer in simplest form:
- $\lim_{x \rightarrow 3} (-x^2 - 2x + 3)$. **2**
 - $\lim_{x \rightarrow -2} \frac{x^2 - x - 6}{x + 2}$. **2**
 - $\lim_{x \rightarrow \infty} \frac{2x^2 - 3x + 7}{5x^2 - 4x + 3}$. **2**
- (b) Differentiate $f(x) = x^2 - 2x$ from first principles. **3**
- (c) The figure below shows the graph of $f(x)$ for $0 \leq x \leq 5$. Sketch the graph of its derivative $f'(x)$. **2**

**End of paper.**

Suggested Solutions

Question 1

(a) i. (2 marks)

✓ [-1] for each mistake.

$$f(x) = 3x^3 + x^2 - 12x + 4$$

$$f'(x) = 9x^2 + 2x - 12$$

ii. (2 marks)

✓ [1] for attempting to apply the chain rule.

✓ [1] for correct final answer. If student gave the final correct answer via expansion, award full marks.

$$y = (5x^2 - 4)^2$$

$$\frac{dy}{dx} = 2 \times 10x (5x^2 - 4)$$

$$= 20x (5x^2 - 4)$$

iii. (2 marks)

✓ [1] for attempting to apply the quotient rule.

✓ [1] for correct final answer. If student gave the final correct answer via product + chain rules, award full marks.

$$f(x) = \frac{3x}{x^2 - 4}$$

$$u = 3x \quad v = x^2 - 4$$

$$u' = 3 \quad v' = 2x$$

$$f'(x) = \frac{vu' - uv'}{v^2}$$

$$= \frac{3(x^2 - 4) - 6x^2}{(x^2 - 4)^2}$$

$$= \frac{3x^2 - 12x - 6x^2}{(x^2 - 4)^2}$$

$$= \frac{-3x^2 - 12}{(x^2 - 4)^2}$$

iv. (3 marks)

✓ [1] for attempting to apply the product rule.

✓ [1] for obtaining $\frac{d}{dx}(x+2)^{\frac{1}{2}} = \frac{1}{2}(x+2)^{-\frac{1}{2}}$.

✓ [1] for correct final answer.

$$y = 5x^3 (x+2)^{\frac{1}{2}}$$

$$u = 5x^3 \quad v = (x+2)^{\frac{1}{2}}$$

$$u' = 15x^2 \quad v' = \frac{1}{2}(x+2)^{-\frac{1}{2}}$$

$$\frac{dy}{dx} = uv' + vu'$$

$$= \frac{1}{2} \cdot 5x^3 (x+2)^{-\frac{1}{2}} + 15x^2 (x+2)^{\frac{1}{2}}$$

$$= \frac{5x^3}{2\sqrt{x+2}} + 15x^2 \sqrt{x+2}$$

(b) (3 marks)

✓ [1] for correct differentiation.

✓ [1] for identifying $\frac{dy}{dx} = 0$ for the tangent to be parallel to the x axis.

✓ [1] for $x = 0, 4$.

$$y = \frac{5}{3}x^3 - 10x^2 + 6$$

$$\frac{dy}{dx} = 5x^2 - 20x$$

The gradient function is $= 0$ when the tangent to the curve is parallel to the x axis.

$$\therefore 5x^2 - 20x = 0$$

$$5x(x - 4) = 0$$

$$\therefore x = 0, 4$$

Question 2

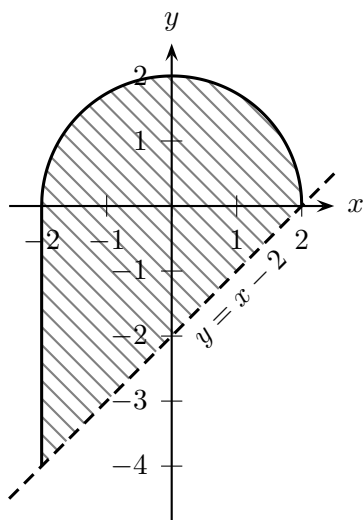
(a) (3 marks)

✓ [1] for correctly sketching semi circle.

✓ [1] for correctly sketching $y = x - 2$ with dashed line.

✓ [1] for correct shading.

$$\{(x, y) : y \leq \sqrt{4 - x^2}\} \cap \{(x, y) : y > x - 2\}$$



(b) (3 marks)

✓ [1] for obtaining

$$(-4, 7) = (9 - 2x, -15 - 2y)$$

✓ [1] each for correct value of x and y obtained from expression.

$$\begin{array}{rcl} A(3, -5) & & -2 \\ & \searrow & \nearrow \\ B(x, y) & & 3 \end{array}$$

$$\begin{aligned} (-4, 7) &= \left(\frac{3(3) - 2x}{1}, \frac{3(-5) - 2y}{1} \right) \\ &= (9 - 2x, -15 - 2y) \\ \begin{array}{l|l} 9 - 2x = -4 & -15 - 2y = 7 \\ 2x = 13 & 2y = -22 \\ x = \frac{13}{2} & y = -11 \end{array} \\ \therefore Q\left(\frac{13}{2}, -11\right) \end{aligned}$$

(c) (3 marks)

$$\begin{cases} x + 2y + 1 = 0 \\ 2x - 3y + 5 = 0 \end{cases} \quad (2.1)$$

$$x(1 + 2k) + y(2 - 3k) + 5k + 1 = 0$$

Substitute $x = 1$ and $y = -3$ to obtain the value of k for the line passing through $(1, -3)$:

$$\begin{aligned} 1 + 2k - 3(2 - 3k) + 5k + 1 &= 0 \\ 16k &= 4 \\ k &= \frac{1}{4} \end{aligned} \quad (2.2)$$

Substitute (2.2) to (2.1)

$$\begin{aligned} \underbrace{x + 2y + 1}_{\times 4} + \underbrace{\frac{1}{4}(2x - 3y + 5)}_{\times 4} &= 0 \\ 4x + 8y + 4 + 2x - 3y + 5 &= 0 \\ 6x + 5y + 9 &= 0 \end{aligned}$$

Question 3

(a) (3 marks)

✓ [1] For attempting to apply the ratio of intercepts.

✓ [1] For each correct final answer. If not fully correct, deduct [1] for each incorrect step.

By the ratio of intercepts,

$$\frac{12}{x+2} = \frac{x+3}{x}$$

Cross multiplying,

$$\begin{aligned} 12x &= (x+2)(x+3) \\ \frac{12x}{-12x} &= \frac{x^2 + 5x + 6}{-12x} \\ x^2 - 7x + 6 &= 0 \\ (x-6)(x-1) &= 0 \\ \therefore x &= 1, 6 \end{aligned}$$

(b) i. (3 marks)

✓ [1] for each reason, up to 2 marks max (subtract [1] for each incorrect reason).

✓ [1] for concluding that the triangles are similar as they are equiangular.

In $\triangle ACB$ & $\triangle LMK$,

- $\angle ACB = \angle KML$
(alternate $\angle$ on $\parallel$ lines AC & ML .)
- $\angle ABC = \angle MKL$
(alternate $\angle$ on $\parallel$ lines AB & KL .)
- $\angle CAB = \angle KLM$
(remaining $\angle$)

$\therefore \triangle ACB \parallel \triangle LMK$ (equiangular). (b) (2 marks)

ii. (2 marks)

$$\begin{aligned}\frac{BC}{KM} &= \frac{AB}{KL} \Rightarrow \frac{2+x}{x+7} = \frac{1}{2} \\ 4+2x &= x+7 \\ CM &= x = 3\end{aligned}$$

✓ [1] for attempting to change base. Penalise if there is no working but only the final answer.

✓ [1] for final answer (1.05). Penalise if it was not rounded to 2 d.p.

(c) i. (3 marks)

✓ [1] for each reason, up to 2 marks max (subtract [1] for each incorrect reason).

✓ [1] for concluding that the triangles are congruent (AAS).

In $\triangle ABE$ and $\triangle DCF$,

• $\angle DFC = \angle AEB$ (data)

• $DC = AB$
(opposite sides of a parallelogram)

• $\angle ABE = \angle DCF$ (corresponding $\angle$ on $\parallel$ lines AB & CD)

$\therefore \triangle ABE \equiv \triangle DCF$ (AAS)

ii. (2 marks)

✓ [1] for each correct reason (subtract [1] for each incorrect reason), up to 2 marks max.

Since $\triangle ABE \equiv \triangle DCF$, $\therefore$ all corresponding sides & angles are equal.

• $AE = DF$
(corresponding sides in $\equiv \triangle$)

• $\angle AEF = \angle EFD = 90^\circ$
(supplementary $\angle$)

• $AE \parallel DF$ as $\angle AEF = \angle EFD$
(corresponding $\angle$ on $\parallel$ lines)

$\therefore AEFD$ is a rectangle – 2 adj. $\angle$ are right angles and 2 opposite sides equal & parallel.

(c) (2 marks)

✓ [1] for taking logs correctly to obtain $x \log_{10} 0.4 \leq 5$.

✓ [1] for final answer $x \geq -12.56$. No need to penalise for 2 d.p. as long as the sign of the inequality is correct.

$$\begin{aligned}\log_9 10 &= \frac{\log_1 010}{\log_1 09} \\ &= \frac{1}{\log_{10} 9} \\ &= 1.05 \text{ (2 d.p.)}\end{aligned}$$

$$0.4^x \leq 5$$

$$\frac{x \log_{10} 0.4}{\div \log_{10} 0.4} \leq \frac{\log_{10} 5}{\div \log_{10} 0.4}$$

$$x \geq \frac{\log_{10} 5}{\log_{10} 0.4} = -1.756$$

(d) (3 marks)

✓ [1] for each correct solution to the quadratic: $x = 1, x = 4$.

✓ [1] for reasoning why $x = 1$ cannot be a solution.

Question 4

(a) i. (1 mark)

$$\log_a 9 = 1.585 \quad \log_a 10 = 1.661$$

$$\log_a 90 = \log_a (9 \times 10)$$

$$= \log_a 9 + \log_a 10$$

$$= 1.585 + 1.661 = 3.246$$

ii. (1 mark)

$$\log_a 0.9 = \log_a \left(\frac{9}{10} \right)$$

$$= \log_a 9 - \log_a 10$$

$$= 1.585 - 1.661 = -0.076$$

$$2 \log_2 (x-1) = \log_2 (3x-3)$$

$$\log_2 (x-1)^2 = \log_2 (3x-3)$$

$$(x-1)^2 = 3(x-1)$$

$$(x-1)^2 - 3(x-1) = 0$$

$$(x-1)((x-1)-3) = 0$$

$$(x-1)(x-4) = 0$$

$$\therefore x = 1, 4$$

However, substituting $x = 1$ into the original equation results in $2 \log_2 0$. The domain of $\log_2 x$ is $x > 0$. Hence the only possible solution is $x = 4$.

Question 5

(c) (2 marks)

(a) i. (2 marks)

✓ [-1] for each incorrect step.

$$\begin{aligned}\lim_{x \rightarrow 3} (-x^2 - 2x + 3) &= -(3^2) - 2(3) + 3 \\ &= -9 - 6 + 3 = -12\end{aligned}$$

ii. (2 marks)

✓ [1] for realising the need to factorise.

✓ [1] for correct final answer.

$$\begin{aligned}\lim_{x \rightarrow -2} \frac{x^2 - x - 6}{x + 2} &= \lim_{x \rightarrow -2} \frac{(x+2)(x-3)}{x+2} \\ &= -2 - 3 = -5\end{aligned}$$

iii. (2 marks)

✓ [1] for dividing the numerator and denominator by x^2 .

✓ [1] for correct final answer.

$$\begin{aligned}\lim_{x \rightarrow \infty} \frac{2x^2 - 3x + 7}{5x^2 - 4x + 3} &= \lim_{x \rightarrow \infty} \frac{2 - \frac{3}{x} + \frac{7}{x^2}}{5 - \frac{4}{x} + \frac{3}{x^2}} \\ &= \frac{2}{5}\end{aligned}$$

(b) (3 marks)

✓ [1] for obtaining

$$f(x+h) = x^2 + 2hx + h^2 - 2x - 2h.$$

✓ [1] for correctly recalling

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}.$$

✓ [1] for final answer.

$$f(x) = x^2 - 2x$$

$$\begin{aligned}f(x+h) &= (x+h)^2 - 2(x+h) \\ &= x^2 + 2hx + h^2 - 2x - 2h\end{aligned}$$

$$\begin{aligned}f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{\cancel{x^2} + 2hx + h^2 - \cancel{2x} - 2h - \cancel{x^2} + \cancel{2x}}{h} \\ &= \lim_{h \rightarrow 0} \frac{\cancel{h}(2x + h - 2)}{\cancel{h}} \\ &= 2x - 2\end{aligned}$$

